

Intro to Statistics

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Intro to Statistics

Statistics is the science of:

- **Collecting** data
- **Organizing** data
- **Presenting** data
- **Analyzing** data

Statistics plays a key role in understanding and interpreting large sets of information (data), and is widely used across various fields.

Types of Data

There are two main types of data:

- **Categorical Data**
- **Numerical Data**

Categorical Data

- represents **labels** or **textual descriptions**.
- This data **cannot be measured numerically**.
 - Examples:
 - **Animal**: Lion, Elephant, Cat
 - **Event**: Eid, Birthday, Wedding
 - **Color**: Red, Blue, Green

Categorical data is used to classify or group information.

Numerical Data

Numerical data represents **quantitative values**.

It is divided into:

- 1. Discrete Data** – countable values
- 2. Continuous Data** – measurable values

Discrete Data

- Represents **countable values**.
- Values are usually **whole numbers** (integers).
- You **cannot have decimals** in discrete data.
 - Examples:
 - Number of students in a class
 - Number of goals in a football match
 - Number of items in a shopping cart

Continuous Data

- Represents **measurable quantities**.
- Values can be **any number, including decimals**.
- Used when data can be measured **with precision**.
 - Examples:
 - Weight: 70.5 kg
 - Height: 165.2 cm
 - Temperature: 36.6°C
 - Time: 2.5 hours

Levels of Measurement

Data is classified into two major levels:

- **Qualitative Data** (Categorical)
- **Quantitative Data** (Numerical)

Each has subtypes depending on how the data behaves.

Qualitative Data

- **Qualitative data** is **text-based** and used to **label or categorize**.
- It is also called **categorical data**.
- It has two types:
 - **Nominal**
 - **Ordinal**

Nominal Data

- **Categories without a specific order**
- You cannot rank or sort them logically
 - Examples:
 - Country: Egypt, Italy, USA
 - Animal: Cat, Dog, Horse
 - Color: Red, Green, Blue

Ordinal Data

- **Categories with a meaningful order or rank**
- Values are ranked, but the **difference between them is not measurable**
 - Examples:
 - Satisfaction level: Poor, Fair, Good, Excellent
 - Movie ratings: ★, ★★, ★★★, ★★★★, ★★★★★
 - Education level: High School, Bachelor's, Master's, PhD

Quantitative Data

- **Quantitative data is numerical.**
 - It represents measurable or countable values.
 - It has two main types:
 - **Interval**
 - **Ratio**

Interval Data

- Numerical values with **equal intervals** between them
 - Has **ordering**, but **no true zero**
 - Zero does **not** mean “nothing”
 - Can have **negative values**
 - Examples:
 - Temperature in Celsius: 0°C does not mean "no temperature"
 - 00:00 o'clock" is not a true absence of time

Ratio Data

- Numerical data with **ordering and a true zero**
 - Zero means the **complete absence** of the quantity
 - Cannot have **negative values**
 - Examples:
 - Height: 0 cm means no height
 - Weight: 0 kg means no weight
 - Money: 0 EGP means no money
- Number of books: 0 books = no books

Types of Statistics

There are two main branches of statistics:

1.Descriptive Statistics

2.Inferential Statistics (*covered later*)

What is Descriptive Statistics?

Descriptive Statistics are used to:

- **Summarize**
- **Organize**
- **Present** the main features of a dataset.

They help you understand **what the data shows** without making predictions.

Example:

You have sales data from a store. Descriptive statistics help you find:

- The **average sales per day**
- The **most sold product**
- How **consistent** sales are over the month

Categories of Descriptive Statistics

1.Measures of Frequency

2.Measures of Central Tendency

3.Measures of Dispersion (Variability)

Measures of Frequency

Definition:

Tells how often each value appears in the dataset.

Example:

Survey Results:

- Apple = 40
- Banana = 25
- Orange = 35 → Most frequent is Apple

Use Case in Business:

Used to identify **popular products**, **most common customer feedback**, or **frequent issues** in customer service.

Helps in **stock planning** and **product prioritization**.

Mean (Average)

Definition:

The total of all values divided by the number of values.

Example:

Daily Sales = 100, 120, 80, 90, 110 → Mean = **100**

Use Case in Business:

To calculate **Average Ticket Size** = Total Sales ÷ Number of Transactions

Helps in **revenue forecasting, sales performance, and customer value analysis.**

Median

Definition:

The middle value when the dataset is ordered.

Example:

Salaries = 10K, 20K, 30K, 40K, 50K → Median = **30K**

Use Case in Business:

Used when the data has **outliers**, like extreme salaries or property prices.

Gives a **realistic central value** for **income analysis**, **market trends**, or **pricing**.

Mode

Definition:

The value that occurs **most frequently** in the dataset.

Example:

Product ratings: 5, 3, 5, 4, 5 $\rightarrow$ Mode = 5

Use Case in Business:

Helps to identify **most common preferences, frequent buying behavior, or repeat customer choices.**

Useful in **product recommendations and inventory planning.**

Measures of Dispersion

These describe how **spread out** the data is.
They help assess **consistency** and **variability** in the dataset.

Definition:

How close or far apart data points are from each other or from the mean.

Why Dispersion Matters

Low dispersion: Data is tightly clustered → **More consistent**

e.g., Test scores: 88, 90, 89 → all close to each other

• **High dispersion:** Data is widely spread → **Less consistent**

e.g., Test scores: 30, 55, 90 → large gap between values

What are Outliers?

- Outliers are values that are **very different** from the rest.

Example:

Salaries: 5K, 6K, 5.5K, 100K → 100K is an **outlier**.

They can affect the **mean** and **range** dramatically.

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1-Range

Definition:

The difference between the **maximum** and **minimum** value.

Example:

Temperature Range = $40^{\circ}\text{C} - 20^{\circ}\text{C} = 20^{\circ}\text{C}$

Use Case in Business:

To understand **data spread**, such as **price variation**, **performance fluctuation**, or **delivery time differences**.

Helps detect **extreme gaps** or **inconsistencies** in operations.

2-Interquartile Range (IQR)

Definition:

Focuses on the **middle 50%** of the data, removes extreme values.

Example:

Salaries: $Q1 = 20K$, $Q3 = 60K \rightarrow IQR = 40K$

Use Case in Business:

Used to find the **most typical customers or employees** while ignoring outliers.

Great for **target market analysis** or **salary benchmarking**.

3-Variance

Definition:

Shows how much the data points **differ** from the mean.

Example:

Sales Variance = High variance means **inconsistent** sales day to day.

Use Case in Business:

To check **consistency of performance**, such as sales, customer spending, or employee performance.
Helps in **process stability** and **quality control**.

4-Standard Deviation

Definition:

Square root of variance, tells how much the data **deviates** from the average.

Example:

Low standard deviation = **stable** daily revenue

High standard deviation = **risky** or **volatile** sales



Use Case in Business:

Used to **measure risk**, **predict performance stability**, and compare **sales variation across regions**.

Critical in **financial modeling**, **inventory control**, and **forecasting**.

Thank You